QUESTION: 9 STACKS USING LINKED LIST

main.c

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```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *next;
};
struct Node *top = NULL
void push(int value) {
    struct Node *newNode;
    newNode = (struct Node *)malloc(sizeof(struct Node));
    if (newNode == NULL) {
        printf("Stack Overflow\n");
        return;
    }
    newNode->data = value;
    newNode->next = top;
    top = newNode;
    printf("%d pushed into stack\n", value);
}
void pop() {
    struct Node *temp;
    if (top == NULL) {
        printf("Stack Underflow\n");
        return;
    }
    temp = top;
```

Output

```
--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter value: 5
5 pushed into stack

--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 2
Deleted element: 5

--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack is Empty
```

main.c

```c
    printf("Deleted element: %d\n", top->data);
    top = top->next;
    free(temp);
}
void display() {
    struct Node *temp;
    if (top == NULL) {
        printf("Stack is Empty\n");
        return;
    }
    temp = top;
    printf("Stack elements are:\n");
    while (temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}
int main() {
    int choice, value;
    while (1) {
        printf("\n--- STACK USING LINKED LIST ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
```

```
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter value: ");
                scanf("%d", &value);
                push(value);
                break;
            case 2:
                pop();
                break;
            case 3:
                display();
                break;
            case 4:
                exit(0);
            default:
                printf("Invalid Choice\n");
        }
    }
    return 0;
}
```